

E. Li Hua and Array

time limit per test: 3 seconds
memory limit per test: 512 megabytes

Li Hua wants to solve a problem about φ — Euler's totient function. Please recall that $\varphi(x) = \sum_{i=1}^x [\gcd(i, x) = 1]$.

He has a sequence $a_1, a_2, \dots, a_n$ and he wants to perform m operations:

- "1 $l\ r$ " ($1 \leq l \leq r \leq n$) — for **each** $x \in [l, r]$, change a_x into $\varphi(a_x)$.
- "2 $l\ r$ " ($1 \leq l \leq r \leq n$) — find out the minimum changes needed to make sure $a_l = a_{l+1} = \dots = a_r$. In each change, he chooses **one** $x \in [l, r]$, change a_x into $\varphi(a_x)$. Each operation of this type is independent, which means the array doesn't actually change.

Suppose you were Li Hua, please solve this problem.

[†] $\gcd(x, y)$ denotes the **greatest common divisor (GCD)** of integers x and y .

[†] The notation $[\text{cond}]$ equals 1 if the condition cond is true, and 0 otherwise.

Input

The first line contains two integers n and m ($1 \leq n, m \leq 10^5$) — the number of elements in the array and the number of operations to process, respectively.

The second line contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 5 \cdot 10^6$) — the elements of the array.

Next m lines, each line contains three integers t_i, l_i, r_i ($t_i \in \{1, 2\}, 1 \leq l_i \leq r_i \leq n$) — the i -th operation.

Output

For each " $2 \mid r$ ", output the answer in an separate line.

Example

input	Copy
5 4 8 1 6 3 7 2 1 5 2 3 4 1 1 3 2 3 4	
output	Copy
10 2 1	

Note

Denote $\varphi^k(x) = \begin{cases} x, & k = 0 \\ \varphi(\varphi^{k-1}(x)), & k > 0 \end{cases}$.

At first, $a = [8, 1, 6, 3, 7]$.

To make sure $a_1 = a_2 = a_3 = a_4 = a_5$, we can change a to $a' = [\varphi^3(8), \varphi^0(1), \varphi^2(6), \varphi^2(3), \varphi^3(7)] = [1, 1, 1, 1, 1]$, using $3 + 0 + 2 + 2 + 3 = 10$ changes.

To make sure $a_3 = a_4$, we can change a to $a' = [\varphi^0(8), \varphi^0(1), \varphi^1(6), \varphi^1(3), \varphi^0(7)] = [8, 1, 2, 2, 7]$, using $0 + 0 + 1 + 1 + 0 = 2$ changes.

After "1 1 3", a is changed to $a = [\varphi^1(8), \varphi^1(1), \varphi^1(6), \varphi^0(3), \varphi^0(7)] = [4, 1, 2, 3, 7]$.

To make sure $a_3 = a_4$, we can change a to $a' = [\varphi^0(4), \varphi^0(1), \varphi^0(2), \varphi^1(3), \varphi^0(7)] = [4, 1, 2, 2, 7]$, using $0 + 0 + 0 + 1 + 0 = 1$ change.

<u>Codeforces Round 864 (Div. 2)</u>
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GNU G++23 14.2 (64 bit, ms² ▼

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